

Data Specification: The "Omics" Engine

1. Robust Data Sources

To ensure "God-Tier" accuracy, the tool integrates the following:

- **Transcriptomics (Bulk):** TCGA (COAD/READ), GEO (GSE39582, GSE33113).
- **Transcriptomics (Single-Cell):** Human Colon Cancer Atlas (scRNA-seq) to see if the PARL-MAVS axis is in tumor cells or immune cells.
- **Proteomics:** CPTAC (Clinical Proteomic Tumor Analysis Consortium) to verify protein-level cleavage.
- **Interactions:** STRING-DB (PPI), BioGRID (Physical interactions), and PDB (Structural motifs).
- **Pharmacology:** DrugBank & CTD (To identify drugs that target this axis).

2. Advanced Analysis Tools

- **Differential Expression:** DESeq2 and edgeR (Gold standards).
- **Survival Modeling:** Cox Proportional Hazards and Random Survival Forests (Machine Learning).
- **Network Science:** Weighted Gene Co-expression Network Analysis (WGCNA).
- **Pathway Enrichment:** GSEA, KEGG, and Reactome via ClusterProfiler .

3. API & Scripting Layer

- **GDC API:** Metadata and clinical fetching.
- **MyGene.info API:** Real-time gene annotation.
- **AlphaFold API:** For predicting the cleavage site of PARL on the MAVS protein structure.